

Yusuf Berk AKÇAY

Curriculum Vitae

Istanbul, Turkey — yusuf.akcay1@std.bogazici.edu.tr — <https://yusufberkakcay52.github.io/>

EDUCATION

Mathematics Major with Double Major in Physics

2023 - 2027 (Expected)

Current Year: 3rd year

Institution: Boğaziçi University, Istanbul, Turkey

Current GPA: 3.98/4

RESEARCH INTEREST

I have a broad interest in modern aspects of mathematical physics, especially in differential geometric foundations of various areas of physics, such as classical mechanics, quantization of classical mechanical systems and statistical mechanics. My Github page provides further details on my academic interests and work.

RESEARCH EXPERIENCE

Skeleta in Complex Projective Line

March 2026 - Now

Supervisor: Asst. Prof. Umut Varolgüneş

- Studied the holomorphic line bundles of complex projective line with Hermitian structures, their sections and associated Liouville vector fields.
- Working to determine the combinatorial types of the skeleta in terms of the associated section, and relate the intersection properties of two skeleta to the overlap estimates between the corresponding quantum coherent states.

Summer Internship in Florence at ISC-CNR

June-August 2026

Supervisor: Prof. Stefano Ruffo

- Analyzed the ergodic and differential geometric aspects of deriving thermodynamics from classical mechanics, in particular the Rugh formalism and its generalizations, and contributed to the relations between these formalisms.
- Investigated various statistical mechanical models, such as the mean-field ϕ^4 model, its variations, and the one-dimensional XY model by implementing symplectic integrators.

INDEPENDENT STUDIES AND ACADEMIC PROJECTS

Reading Program on Geometry of Quantization

September 2025 - April 2026

Supervisor: Asst. Prof. Umut Varolgüneş

- Learned about symplectic geometry and explored the applications of symplectic geometric ideas to physics, such as formulation of the classical mechanics on manifolds, the WKB method to obtain approximate solutions of the Schrödinger equation, Maslov correction and geometric quantization of symplectic manifolds.

Course Project, PHYS 177: Computational Introduction to Dynamical Systems

July 2025

- Prepared a report on solvable structures and their relation to the Arnold-Liouville theorem.

PHYS 491: Introduction to Research in Physics I

September-December 2024

Supervisor: Asst. Prof. Ilmar Gahramanov

- Studied the Dunkl deformed quantum mechanics with an emphasis on the deformed Pauli oscillator, and applied the Lewis-Riesenfeld Invariant to obtain exact solutions of the Schrödinger equation.
- Delivered a presentation on the topic to an undergraduate level audience.
- Investigated the possible supersymmetric extensions of the topic.

SCHOOLS AND WORKSHOPS

Istanbul Stringy Meeting

Boğaziçi University, Istanbul, Turkey

2026

Summer School on the Digital Archive of Exact Solutions in General Relativity

Istanbul Technical University, Istanbul

2025

37th National Mathematics Symposium

Akdeniz University, Antalya, Turkey

2025

Grad. and Undergrad. Summer Math School
 Nesin Mathematics Village, Izmir, Turkey 2025

- Category Theory
- Conformal Field Theory
- Selected Topics in Modern Physics

COSMOVERSE@ISTANBUL 2025
 Istanbul Technical University, Istanbul, Turkey 2025

Istanbul Stringy Meeting
 Boğaziçi University, Istanbul, Turkey 2024

TALKS

Directed Reading Program 2026
 N=4 Super Yang-Mills Theory 2026

- Introduction to Quantum Field Theory, with an emphasis on renormalization, Yang-Mills theory, supersymmetry, and the construction of the N=4 supersymmetric Yang-Mills Lagrangian

Boğaziçi Math Grad Seminars
 WKB Approximation and Geometric Quantization 2026

- Connection between the WKB approximation and the framework of Geometric Quantization

Workshop on Symmetry and Geometry in Memory of Yavuz Nutku
 Hydrodynamic Systems with Two Conserved Quantities 2026

- Integrability and Bi-Hamiltonian structures of d-Boussinesq and Benney-Lax equations

HONOURS & AWARDS

- TÜBİTAK 2205 Undergraduate Scholarship — 2023 - Present
- Ranked Top $\sim 0.01\%$ (out of ~ 3.5 million candidates), YKS Turkish National University Entrance Exam — 2023
- Bronze Medal, TÜBİTAK National Physics Olympiad — 2021

SKILLS

Languages: **Turkish**(native), **English**(fluent)
 Computational Tools: **Python**, **Mathematica**, **LaTeX**

EXTRACURRICULAR ACTIVITIES

- Club basketball player for 2 years.
- Certified scout for 4 years.
- Organized voluntary lectures for high school students.